

EM 520 – Engineering Value & Economic Analysis

ONLINE AND FACE2FACE

COLLEGE OF TECHNOLOGY

SCHOOL OF ENGINEERING TECHNOLOGY

Credit Hours: 3

Course Description

Understanding and applying engineering economic principles for financial analysis and management of people, projects, engineering/business processes, and organizational units. Utilize financial principles for investment decision making. Exercise value engineering practices to improve the total value provided to customers.

Textbook(s)

Required: Applied economic analysis for technologists engineers and managers. -- Michael S. Bowman
Upper Saddle River NJ: Prentice Hall., 2nd. ed. 2003. ISBN: 0-13-094511-0

Recommended: Student study guide to accompany applied economic analysis for technologists engineers and managers. -- Michael S. Bowman
Upper Saddle River NJ: Prentice Hall., 2nd. ed 2003. ISBN: 0-13-044990-3

Course Objectives

1. Formulate steps for an engineering economic analysis of the engineering design process.
2. Identify and quantify cost elements throughout the product life cycle.
3. Determine an engineering project's worth.
4. Compare engineering project alternatives.
5. Conduct breakeven analysis.
6. Perform replacement analysis for industrial machinery with quality and product design considerations.
7. Utilize depreciation allowances and income taxes in engineering economic analysis.
8. Understand value engineering concepts and principles and apply them to a common product.

Grading Policy

Grading Scale

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

A	93-100%
A-	90-92%
B+	87-89%
B	83-86%
B-	80-82%
C+	77-79%
C	73-76%
C-	70-72%
E	below 70

Final course grades will be based as shown above, but may be subject to adjustment based on the judgment of your professor.

Course Schedule

TENTATIVE Course Outline and Schedule

Unit	Week(s)	Chapter(s)	Topic(s)
Unit 1	1	1, 2, 3	Intro to Engineering Economics, Intro to Financial Statements: Income Statement, Balance Sheet, Cash Flow Statement, Depreciation, Inventory, and Financial Ratios
Unit 2	2	4, 5	Time Value of Money, Future Value, Gradient Function, Perpetuity, Annuities, Deferred Annuities
Unit 3	3	6-7	Applications of the Time Value of Money, Uncertainty, Probability, and Risk
Unit 4	4	8-9	Four Analytical Methods: Internal Rate of Return, Present Worth, Annual Amount, Future Worth
Unit 5	5	10, 11	Breakeven Analysis, Minimum Cost Analysis Concepts and Techniques, Value Engineering Concepts

Unit 6	6	12	Replacement Analysis and Equipment Life, Taxes, ROI and Cash Flow, Value Engineering Concepts
Unit 7	7	13	Costs and Quality; Value Engineering Applications;
Unit 8	7 & 8 (half)	All	FINAL EXAM & FINAL PROJECT (Case Study)

General Policy

A. Students are highly recommended to contact their professors about their problems about the course or any issues in the class or via email.

B. Academic Dishonesty: The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "E" for the course in which the activity occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work, papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or taking credit for another's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

Our managerial approach in Engineering Management program regarding to minor occurrences of academic dishonesty is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an example of the problem with the rest of the class, as a learning experience for the whole class. The consequences may be more severe (such as failure of the assignment or the entire course), and without a warning, for second and/or major occurrences of academic dishonesty.

C. Writing Style and Format: All students must use APA (American Psychological Association) style in all their written assignments (e.g. paper). However there are some professors who emphasize more on using APA. Students are encouraged to have a copy of the last edition (6th) of the APA manual to use it during their studies.

For more information about traditional and electronic referencing guidelines for quoting, paraphrasing, and referencing information that you use, to help avoid Academic Dishonesty problems, as mentioned above. See <http://www.apa.org/journals/acorner.html#pubmanual> for more specific information and examples. Although the newest APA edition is 6th, you may still use 5th edition.

D. Incomplete Final Grades (I): will be given only in accordance with EMU's guidelines, which are stated in the 2005-2007 Graduate Catalog (p.19), at <http://www.emich.edu/public/catalogs/2005-2007/grad/20051gfront.pdf>, as follows: **I (Incomplete) Grades** - An I grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course (50% or more) merits a grade of B or better.

E. Student and Exchange Visitor Information Service (SEVIS): For students who hold an F or J visa, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), King Hall, any change in name, current address of residence, academic status (full or part-

time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government -- Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services -- SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003). OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

Due to SEVIS regulations, international students studying in the US on an F-1 or J-1 visa are allowed to take only one online class per semester, if face-to-face classes are available. Exceptions to this policy (when face-to-face classes are not available) must be approved prior to registration for any excess online classes by the Office of International Students (OIS), your online class professor, and your academic advisor. In these cases, you must meet in person with the professors of your excess online classes at least 4 times during the semester.

Miscellaneous

1. The instructor controls and monitor the threaded discussions. In this section, you may just use for anything related to the course and don't use it for instance selling your car, and etc.
2. Due to high volumes of e-mails, instructors are not able to respond your e-mails in a while and it takes 2-3 days.
3. In order to be successful in each course , at least 10-12 hours per week is needed and you need to practice through your assignments problems.

Participation Guidelines and Requirements

Here are some guidelines and requirements for making things go smoothly.

A. Access to Course.

Our course consists of 8 units, which are scheduled from Mondays through Sundays. All units will be scheduled across two weeks . The course will be accessible in Fridays prior to the class and will be due on Sundays nights at 11:59 PM. Mostly each unit takes about two weeks.

B. Participation.

In any course, students seek interaction with other students and with the instructor. If this class is offered as Hybrid class then there will be some traditional classroom settings and that will provide a great opportunity for a face to face communication. However, if the class is offered as an online class then we need to take a different approach. As there won't be any required face-to-face meetings, such interaction will take place through virtual meetings, either through email, phone calls, conference calls, chat, Discussion group, IM, tweeter, etc. Such virtual communications thus becomes very important to effective learning online. As such, good class participation is essential and will account for a significant part of your grade. [Plan to contribute to every discussion topic with a "quality" contribution -- i.e. a response that expresses clear thinking, and that is relevant to the discussion. This usually involves more than just "I agree."]

C. Online Conduct.

For hybrid or online classes, I believe that discussion on the online conduct is very important. Nevertheless, as we have students from various countries and culture I will rather state for the sake of clarity, please realize that the University and I cannot and will not be responsible for the content of any personal messages that are sent from one student directly to another student using the online email delivery system. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using email, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.

D. Email to Classmates.

Students are free to use the "email the class" feature on the course web site. Try to keep the messages sent to the whole class related to course content. Also, when emailing the class, generally it is best NOT to email phone numbers or addresses unless it is being emailed directly to a specific individual or the instructor. This helps maintain privacy.

E. Email to Instructor.

For EM 520 course-related submissions and communications, please send all email directly to Ali Dehghan at adehghan@emich.edu. In urgent cases, you may Cc to Dr. Ahmed at mahmed6@emich.edu. Also, in the subject line of your email messages, include the: **initials of your full name**, **course number**, and the **subject of the message** (such as the name of an assignment that is attached to your email); for example, Subject: **XYZ520Unit1**.

I can not be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 520-related email before 12 noon on several days during the week. Due the amount of email I will receive from students, as well as other work and travel commitments, I may not be able to respond to emails or confirm receipt of submitted assignments more than once or twice a week--although I will try to do so more frequently. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor a fellow student may be able to respond quickly enough to help you. Also, it may take a day or two before you receive an emailed confirmation from me that I received your submitted assignment. Typically, the confirmation will read: "Got it, Ali." or "Got it, Dr. Sohail."

F. Email Attachments.

There are so many different word processing programs that email attachments can become a problem in online courses if there is no standard policy set for them. It is preferred that you send any paper that is an attachment to an email in MS Word. Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format are also acceptable. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message--although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2007 or MS Project 2003 format. You may also send single or multiple files in .zip compressed format.

The file name of your attachment should be similar to the subject line of your email message, including the: **initials of your full name**, **course number**, and the **name of the attachment with its file type**; for example, the file name of an attached assignment would look like: **XYZ520unit1.pdf**.

G. Grammar, Spelling, Writing Style, and Format.

We all tend to become somewhat lax when posting responses to threaded discussions or when sending email messages. Grammar and spelling is important to me. I am not a fanatic about it, but I would appreciate your composing threaded discussion responses or email messages in a word processing program where you can check spelling, edit, and revise your writing. Then you can cut and paste the work into the threaded discussion or email message. You can do the same for other coursework and thereby help with NOT tying up Internet Service Provider lines.

Good grammar, spelling, writing style, and APA format is required for formal papers, per SET Department guidelines.

H. Assignment Delivery and Technical Problems.

Assignments that are submitted directly via the EMU Online e-College course system (such as threaded discussion responses, attachments to threaded discussion responses, quizzes, exams) must be submitted before midnight on Sunday night of their due date (by Sunday 11:59 PM ET US), and **will not be accepted late**. I will reluctantly accept late assignments of other types (those types of assignments that are supposed to be emailed directly to me--which are not supposed to be submitted through an e-College course tool), but they will be penalized **20% per day late (Monday 12:00 AM to 11:59 PM ET US is the definition of one day late)**. Since you will be working on a team, others will be depending upon your getting your work done on time. I also expect you to get it done on time. All final unit coursework is due according to what is posted in the assignment tab for that last unit and will not be accepted late. Because of the nature of the EMU Online threaded discussion and examination and administration system, threaded discussions and exams taken or submitted late can not and will not be accepted.

I encourage you to make hard copies of your assignments and of your communications with me. Computer crashes and disasters are bound to happen. Keep in mind that I do not view ISP problems as an excuse for turning work in late. At the first sign of problems with email, contact your ISP immediately. If your problems are not cleared up within 12 hours, send in the assignment in one of the following ways:

1. Fax it to Dr. Sohail at 734.487.8755
begin_of_the_skype_highlighting 734.487.8755 end_of_the_skype_highlighting (EMU).
2. Overnight express hard copy and on disk.
3. Make arrangements over the phone with Ali Dehghan or in urgent cases with Dr. Sohail.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public Library. My guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate.

I. Quizzes and Exams.

If we have quizzes and exams in this course, they will usually be timed, based on a easy standard of 120 seconds (two minutes) per question. The typical standard for professional-level exams, such as certification or licensure exams, is about 60 to 90 seconds per question. What makes it even easier for you, is that your quizzes and exams in our course are very focused on specific information from sections of specific books--whereas certification and licensure exams cover a very wide body of professional knowledge, covered by numerous books and other material.

Like those professional exams, your quizzes and exams in this class are open-book (of course)--and, like those professional exams (and regular classroom exams), you should prepare for taking your quizzes and exams in advance, by reading the material to be covered, taking notes, and applying other test-preparation studying techniques.

J. Personal Involvement and Time.

If this is an online course then I would point out that students often have the mis-perception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier!

Whether its a hybrid or an online course, be aware of the general expectation of professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so I expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and

online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10-15 (or more) hours of your time each week.

Remember, this is a course for the Engineering Management program. Working well together in teams, and making sure that you get your assignments done for your team are essential for project success.

Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Many of our EM students have very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program can not organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. As such, I will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.